AP BIOLOGY TOPIC 7.9

Phylogeny: Phylogenetic Trees, Cladograms & Clades

UNIT 7: NATURAL SELECTION (13–20% EXAM WEIGHT)

EVO-3.B | **EVO-3.C** | **SYNAPOMORPHIES** | **MAXIMUM PARSIMONY**

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 7.9 (PHYLOGENY: PHYLOGENETIC TREES, CLADOGRAMS & CLADES)

CED Core Vocabulary

Phylogeny — The evolutionary history and line of descent of a species or group of related organisms.

Phylogenetic Tree — Branch lengths are frequently calibrated to represent geological time (via fossils/molecular clocks) or the amount of genetic change (nucleotide substitutions).

Nodes (Branch Points) — Represent speciation events and the Most Recent Common Ancestor (MRCA) of the descendant lineages.

Sister Taxa — Two lineages that diverge from the exact same immediate node — they share an MRCA not shared with any other group and are each other's closest relatives.

Phylogenetic Trees vs. Cladograms — Cladogram: Shows branching order (relative relatedness) only based on shared derived characters; branch lengths do not represent time or amount of genetic change.

Key Structural Components — Root: The basal lineage representing the oldest common ancestor of all taxa on the tree.

Tips (Terminal Taxa) — Extant (living) or extinct species at the ends of branches.

Outgroup — A distantly related lineage that diverged before the ingroup; serves as a reference point to distinguish ancestral vs. derived traits.

§2. 1. ANATOMY OF PHYLOGENETIC TREES & CLADOGRAMS

Evolutionary Hypotheses

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

1. Anatomy of Phylogenetic Trees & Cladograms ➔ **Phylogeny** ➔ **Phylogenetic Trees vs. Cladograms** ➔ **Key Structural Components**

- **Phylogeny:** The evolutionary history and line of descent of a species or group of related organisms.
- **Phylogenetic Trees vs. Cladograms:**
  - **Cladogram:** Shows **branching order (relative relatedness) only** based on shared derived characters; branch lengths do *not* represent time or amount of genetic change.
  - **Phylogenetic Tree:** Branch lengths are frequently calibrated to represent **geological time** (via fossils/molecular clocks) or the **amount of genetic change** (nucleotide substitutions).
- **Key Structural Components:**
  - **Root:** The basal lineage representing the **oldest common ancestor** of all taxa on the tree.
  - **Nodes (Branch Points):** Represent **speciation events** and the **Most Recent Common Ancestor (MRCA)** of the descendant lineages.
  - **Tips (Terminal Taxa):** Extant (living) or extinct species at the ends of branches.
  - **Sister Taxa:** Two lineages that diverge from the **exact same immediate node** — they share an MRCA not shared with any other group and are each other's closest relatives.
  - **Outgroup:** A distantly related lineage that diverged *before* the ingroup; serves as a reference point to distinguish ancestral vs. derived traits.

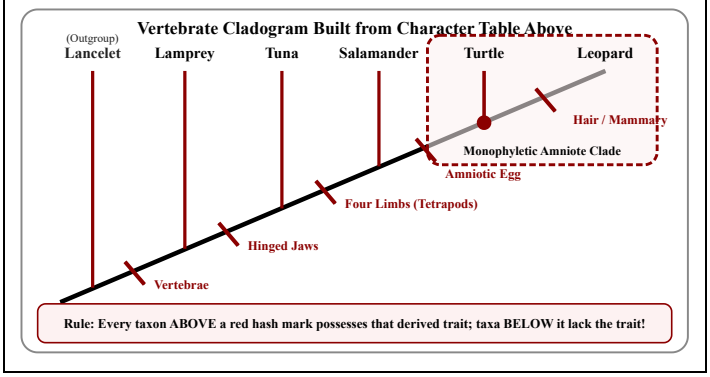
[MECHANISTIC NOTE] **Tree = Time/Change + Relationships | Cladogram = Branching Relationships Only.**

[COMMON AP EXAM FRQ TRAP] **Rotating Around a Node Does NOT Change Relationships!** Never read across the tips from left to right to judge relatedness — **ONLY look at how recently two taxa share a common node (MRCA)**!

§5. HOW TO BUILD A CLADOGRAM FROM A CHARACTER MATRIX

FRQ Construction Skill

TAXON	VERTEBRAE	HINGED JAWS	FOUR LIMBS	AMNIOTIC EGG	HAIR / FUR
Lancelet (Outgroup)	0	0	0	0	0
Lamprey	**1**	0	0	0	0
Tuna (Fish)	**1**	**1**	0	0	0
Salamander	**1**	**1**	**1**	0	0
Turtle	**1**	**1**	**1**	**1**	0
Leopard	**1**	**1**	**1**	**1**	**1**



[AP EXAM FRQ SCORING STRATEGY] **3-Step Recipe for AP FRQ Cladogram Drawing:**

- **1. Identify the Outgroup:** The species with all **0**s (or highest number of amino acid differences) goes on the first basal branch closest to the root.
- **2. Group Sister Taxa:** Place the two species with the most shared derived traits (**1**s) or fewest DNA differences together on the most recent split.
- **3. Place Character Hash Marks:** Place each trait hash mark on the internode *just below* the node uniting all species that possess that trait!

§3. 2. SYNAPOMORPHIES, CLADES & GROUPINGS

Monophyletic Clades

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

2. Synapomorphies, Clades & Groupings ➔
Shared Derived Character (Synapomorphy) ➔
Three Types of Taxonomic Groupings

- **Shared Ancestral Character (Symplesiomorphy):** A trait that originated in an ancestor *prior* to the clade of interest (e.g., backbone in mammals — shared with all vertebrates).
- **Shared Derived Character (Synapomorphy):** An evolutionary novelty unique to a specific clade, inherited from its MRCA (e.g., hair & mammary glands in mammals). Traits can be **gained or secondarily lost** (e.g., legs lost in snakes) over evolution.
- **Three Types of Taxonomic Groupings:**
  - **Monophyletic Group (True Clade):** Includes a single common ancestor (node) and **ALL** of its descendants ("one-snip test").
  - **Paraphyletic Group:** Includes a common ancestor and *some*, but not all, of its descendants (e.g., traditional "Reptilia" excluding birds).
  - **Polyphyletic Group:** Groups distantly related organisms without their common ancestor (often based on convergent analogous traits, e.g., "warm-blooded animals" = birds + mammals).

[CORE AXIOM / REMEMBER] **Shared derived traits (synapomorphies) = evolutionary "clues" linking species into clades.**

§4. 3. BUILDING TREES: MOLECULAR DATA & MAXIMUM PARSIMONY

CED EVO-3.C

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

3. Building Trees ➔ **Maximum Parsimony (Occam's Razor)** ➔
Why Molecular Data Beats Morphology Alone ➔
Phylogenetic Trees Are Hypotheses

- **Maximum Parsimony (Occam's Razor):** When choosing among competing phylogenetic trees, scientists prefer the tree that requires the **fewest total evolutionary events** (fewest mutations or character gains/losses).
- **Why Molecular Data Beats Morphology Alone:**
  - Morphological traits can be misleading due to **convergent evolution (analogous structures)** or extreme divergence.
  - **DNA, RNA, and protein sequences** provide thousands of objective nucleotide positions and reveal true evolutionary relationships even when morphology converges.
- **Phylogenetic Trees Are Hypotheses:** Trees are constantly revised as new fossils or genome sequences are discovered!

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 7.9 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 7.9: Phylogeny: Phylogenetic Trees, Cladograms & Clades ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 1, 2 & 3 (Artificial Selection, Hardy-Weinberg Mathematical Modeling & BLAST Bioinformatics): Comparing allele frequency shifts across generations under directional selection vs. small-population genetic drift simulations, and constructing maximum-parsimony cladograms from BLAST percentage sequence identity matrices (Cytochrome c / β-globin). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Golden Rule: Hardy-Weinberg Equilibrium (HWE): Allele frequencies $p + q = 1$; Genotype frequencies p^2 (homozygous dominant) + $2pq$ (heterozygous carriers) + q^2 (homozygous recessive) = 1. Golden Calculation Sequence: ALWAYS begin with the recessive phenotype frequency (q^2), take $\sqrt{(q^2)}$ to find q, compute $p = 1 - q$, and then calculate $2pq$ (heterozygote carrier frequency). Never take the square root of the dominant phenotype frequency ($p^2 + 2pq$)! **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Evolutionary & Biomedical Disruption Case Study: Incomplete antibiotic regimens or agricultural antibiotic overuse exert strong directional selection on bacterial populations (e.g., MRSA or MDR-Tuberculosis): susceptible cells die while rare, pre-existing resistant mutants (arising via random mutation or R-plasmid conjugation) survive and reproduce. In conservation biology, severe population bottlenecks (e.g., Northern Elephant Seals, African Cheetahs) drastically erode genetic diversity via genetic drift, increasing homozygosity for deleterious recessive alleles and vulnerability to novel pathogens.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 7.9 to prove why the claim holds.

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